MTH319 2023S2 Exam

Feel free to contact me at maths_and_programming@outlook.sg or @geodome on Telegram if you need tuition.

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Donaldson Tan

SINGAPORE UNIVERSITY OF SOCIAL SCIENCES

MTH319

Examination - July Semester 2023

Mathematical Methods I

Friday, 17 November 2023 **04:00 pm - 06:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **FIVE (5)** questions and comprises **FIVE (5)** printed pages (including cover page).

2. You must answer **ALL** questions.

3. All answers must be written in the answer book. Marks will only be awarded if FULL working is shown.

4. This is an **Open-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

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(Full marks: 100)

Question 1

Given a system of linear equations as:

$$\begin{aligned} 4x - 2y - 5z &= 7 \\ x - y - 2z &= 2 \\ x + y + z &= 1. \end{aligned}$$

Question 1a

Using matrix row operations, show that its solution is :

$$z = -\frac{1}{3}(2y + 1) = 2x - 3.$$

Describe the system in terms of its linear independence , consistency and rank. What does the solution represent?

(12 marks)

Question 1b

Show that the point $(1, 1, -1)$ is a solution given in Question 1a. Hence, determine the vector equation:

$$\vec{r} = \vec{u} + t\hat{v}$$

which also represents the same solution, where v is a unit vector.

(8 marks)

Question 2

Figure Q2 below shows 4 data points.

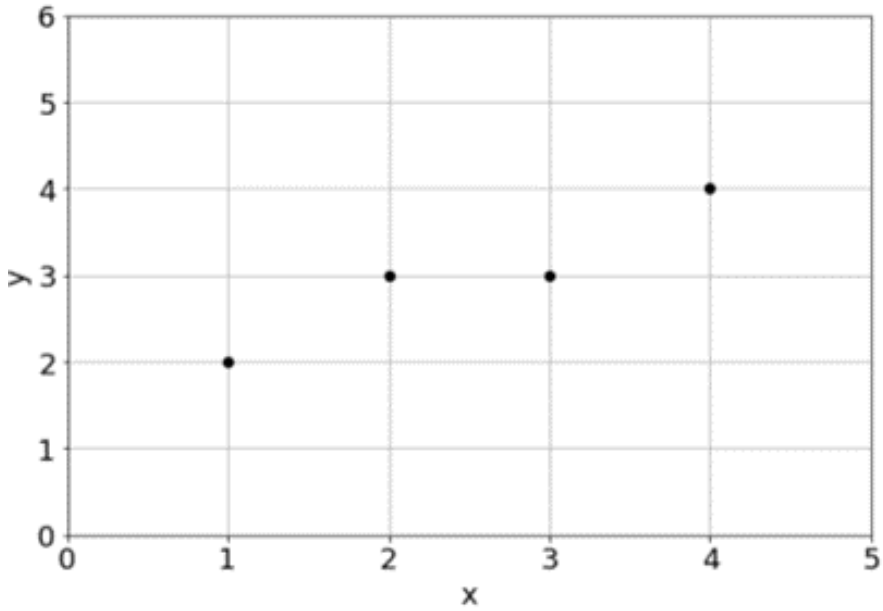

Figure Q2

Question 2a

Using linear regression by minimising the sum of squared errors, show that the equation of the best-fit line for the above dataset is :

$$y(x) = 0.6x + 1.5.$$

(12 marks)

Question 2b

Calculate the sum of squared errors for the best-fit line given in Question 2a.

(5 marks)

Question 2c

If a new data point given by $(5, 4.5)$ is added, how would the linear fit change? Explain briefly.

(3 marks)

Question 3

Given the matrix A below:

$$A = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -2 & 1 \\ 0 & 1 & -2 \end{pmatrix}$$

Compute the eigendecomposition of A. For each eigenvalue, describe its eigenspace.

(15 marks)

Question 4

A light bulb generates heat at $q(t)$ Watts when it is turned on and loses the heat through the glass surface of the bulb exposed to the surroundings at temperature T_s. The glass body has mass m, specific heat capacity c and exposed area A to the surroundings.

Question 4a

The light bulb is turned on at $t = 0$ after being off for a long time. By applying the conservation of energy to the glass body , derive the mathematical model that governs the heat transfer phenomenon and show that it results in the initial value problem:

$$T' + \lambda T = \frac{q(t)}{mc} + \lambda T_s, \;\; T(0) = T_S$$

where $T(t)$ represents the temperature function of the glass body and U is the overall heat transfer coefficient. The parameter $\lambda = UA/(mc)$. State your assumptions clearly.

(10 marks)

Question 4b

If the heat generation by the light bulb has the function:

$$q(t) = Q(1 - e^{-t}) \quad \text{(Watts)}$$

where Q is a constant, determine the temperature function of the class body.

(15 marks)

Question 5

A matrix is defined as :

$$A = \begin{bmatrix} 1 & -4 \\ 4 & 9 \end{bmatrix}.$$

Question 5a

Show that the Jordan matrix decomposition of A is :

$$A = \begin{bmatrix} 1 & 0 \\ -1 & -\frac{1}{4} \end{bmatrix} \begin{bmatrix} 5 & 1 \\ 0 & 5 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -4 & -4 \end{bmatrix}.$$

(12 marks)

Question 5b

Hence , solve the system of ODEs given below. What happens to the solutions as time t approaches infinity?

$$\begin{aligned} x' &= x - 4y \\ y' &= 4x + 9y \end{aligned} \quad , x(t=0) = 0, y(t=0) = 1$$

(8 marks)

----- END OF PAPER -----

1(a) $z = -\frac{1}{3}(2y+1)$

$3z = -(2y+1)$

$2y + 3z = -1 \quad -(1)$

$z = 2x - 3$

$-2x + z = -3 \quad -(2)$

row operations must produce equations (1) and (2)

$$\left[\begin{array}{ccc|c} 4 & -2 & -5 & 7 \\ 1 & -1 & -2 & 2 \\ 1 & 1 & 1 & 1 \end{array}\right] \xrightarrow[r_3 = 2r_3 - r_1]{r_2 = r_3 - r_2} \left[\begin{array}{ccc|c} 4 & -2 & -5 & 7 \\ 0 & 2 & 3 & -1 \\ -2 & 4 & 7 & -5 \end{array}\right] \xrightarrow{r_3 = r_3 - 2r_2} \left[\begin{array}{ccc|c} 4 & -2 & -5 & 7 \\ 0 & 2 & 3 & -1 \\ -2 & 0 & 1 & -3 \end{array}\right]$$

$\Rightarrow 2y + 3z = -1 \Leftrightarrow z = -\frac{1}{3}(2y+1)$

$\Rightarrow -2x + z = -3 \Leftrightarrow z = 2x - 3$

$$\left[\begin{array}{ccc|c} 4 & -2 & -5 & 7 \\ 0 & 2 & 3 & -1 \\ -2 & 0 & 1 & -3 \end{array}\right] \xrightarrow{r_3 = r_1 + 2r_3} \left[\begin{array}{ccc|c} 4 & -2 & -5 & 7 \\ 0 & 2 & 3 & -1 \\ 0 & -2 & -3 & 1 \end{array}\right] \xrightarrow{r_3 = r_2 + r_3} \left[\begin{array}{ccc|c} 4 & -2 & 5 & 7 \\ 0 & 2 & 3 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

There are 2 pivots, so Rank = 2

The system is consistent

The number of pivots is less than 3, so there is no linear independence

There is 1 free variable, so there exists many solutions

1 (b) $-\frac{1}{3}(2y+1) = 2x-3$

$$2y+1 = 9-6x$$
$$2y = 8-6x$$
$$y = 4-3x$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ 4-3x \\ 2x-3 \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \\ -3 \end{bmatrix} + x \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix}$$

let $x=1$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \\ -3 \end{bmatrix} + \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

$\therefore$ $(1,1,-1)$ is a solution given in question 1(a)

$$\Rightarrow \underset{\sim}{u} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

$$\underset{\sim}{m} = \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix} \qquad |\underset{\sim}{m}| = \sqrt{1+3^2+2^2} = \sqrt{14}.$$

$$\Rightarrow \underset{\sim}{v} = \frac{1}{|\underset{\sim}{m}|}\underset{\sim}{m} = \frac{1}{\sqrt{14}} \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix}$$

$$\therefore \underset{\sim}{r}(t) = \underset{\sim}{u} + t\underset{\sim}{v}$$

$$= \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} + \frac{t}{\sqrt{14}} \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix}$$

2(a) $\begin{bmatrix} m \\ c \end{bmatrix} = (A^TA)^{-1} A^T Y$ where $Y = \begin{bmatrix} 2 \\ 3 \\ 3 \\ 4 \end{bmatrix}$, $A = \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \\ 4 & 1 \end{bmatrix}$

$A^TA = \begin{bmatrix} 30 & 10 \\ 10 & 4 \end{bmatrix}$ $\quad (A^TA)^{-1} = \frac{1}{30(4)-10^2} \begin{bmatrix} 4 & -10 \\ -10 & 30 \end{bmatrix} = \frac{1}{10} \begin{bmatrix} 2 & -5 \\ -5 & 15 \end{bmatrix}$

$A^TY = \begin{bmatrix} 33 \\ 12 \end{bmatrix}$

$$\begin{bmatrix} m \\ c \end{bmatrix} = (A^TA)^{-1}(A^TY) = \frac{1}{10} \begin{bmatrix} 2 & -5 \\ -5 & 15 \end{bmatrix} \begin{bmatrix} 33 \\ 12 \end{bmatrix} = \frac{3}{10} \begin{bmatrix} 2 \\ 5 \end{bmatrix}$$

$\therefore y(x) = mx + c = 0.6x + 1.5$

(b)

x	y	$y(x)$	$y - y(x)$	$(y - y(x))^2$
1	2	2.1	-0.1	0.01
2	3	2.7	0.3	0.09
3	3	3.3	-0.3	0.09
4	4	3.9	0.1	0.01

sum of squared errors $= 0.01 + 0.09 + 0.09 + 0.01 = 0.20$

(c) $y(5) = 0.6(5) + 1.5 = 4.5$

$\Rightarrow (5, 4.5)$ lies on the line of regression

Hence, the linear fit doesn't change

3. let $\det(A-\lambda I)=0$

$$\begin{vmatrix} -1-\lambda & 0 & 0 \\ 0 & -2-\lambda & 1 \\ 0 & 1 & -2-\lambda \end{vmatrix} = 0$$

$$(-1-\lambda)[(-2-\lambda)^2-1] = 0$$
$$(\lambda+1)[(\lambda+2)^2-1] = 0$$
$$(\lambda+1)(\lambda^2+4\lambda+3)=0$$
$$(\lambda+1)(\lambda+1)(\lambda+3)=0$$
$$\therefore \lambda_{1,2}=-1$$
$$\therefore \lambda_{2,1}=-3$$

case $\lambda=\lambda_{1,2}=-1$:

$$\left[\begin{array}{ccc|c} 0 & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 1 & -1 & 0 \end{array}\right] \xrightarrow{r_2=r_2+r_3} \left[\begin{array}{ccc|c} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \end{array}\right]$$

$$\Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_3 \\ x_3 \end{bmatrix} = x_1\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + x_3\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

eigenspace of $\lambda_{1,2}$ = span$\left\{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}\right\}$

case $\lambda=\lambda_{2,1}=-3$:

$$\left[\begin{array}{ccc|c} 2 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \end{array}\right] \xrightarrow[r_3=r_3-r_2]{r_1=\frac{1}{2}r_1} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ -x_3 \\ x_3 \end{bmatrix} = x_3\begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$$

eigenspace for $\lambda_{2,1}$ = span$\left\{\begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}\right\}$

$\therefore A=PDP^{-1}$ where $P=\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 1 & 1 \end{bmatrix}$, $D=\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -3 \end{bmatrix}$

4a. By law of Conservation of Energy,

heat generated - heat loss = heat accumulated

heat generated = $q(t)$

heat loss = $UA(T - T_s)$ [By convection]

heat accumulated = mcT' [assuming no state change or chemical reaction]

$$q(t) - UA(T - T_s) = mcT'$$

$$\frac{q(t)}{mc} - \frac{UA}{mc}(T - T_s) = T'$$

$$\frac{q(t)}{mc} - \lambda(T - T_s) = T'$$

$$\frac{q(t)}{mc} - \lambda T + \lambda T_s = T'$$

$$\therefore T' + \lambda T = \frac{q(t)}{mc} + \lambda T_s$$

$T(0) = T_s$ because bulb is at ambient temperature before it was switched on.

(b) $q(t) = Q(1 - e^{-t})$

$$T' + \lambda T = \frac{Q(1-e^{-t})}{mc} + \lambda T_s$$

use integrating factor $= e^{\int \lambda dt} = e^{\lambda t}$

$$e^{\lambda t}[T' + \lambda T] = e^{\lambda t}\left[\frac{Q(1-e^{-t})}{mc} + \lambda T_s\right]$$

$$e^{\lambda t}T' + \lambda e^{\lambda t}T = \frac{Q}{mc}e^{\lambda t} - \frac{Q}{mc}e^{(\lambda-1)t} + \lambda T_s e^{\lambda t}$$

$$\frac{d}{dt}(e^{\lambda t}T) = \frac{Q}{mc}e^{\lambda t} - \frac{Q}{mc}e^{(\lambda-1)t} + \lambda T_s e^{\lambda t}$$

$$e^{\lambda t}T = \int \frac{Q}{mc}e^{\lambda t} - \frac{Q}{mc}e^{(\lambda-1)t} + \lambda T_s e^{\lambda t}\, dt$$

$$= \frac{Q}{\lambda mc}e^{\lambda t} - \frac{Q}{(\lambda-1)mc}e^{(\lambda-1)t} + T_s e^{\lambda t} + C$$ where C is a constant

$$T(t) = \frac{Q}{\lambda mc} - \frac{Q}{(\lambda-1)mc}e^{-t} + T_s + Ce^{-\lambda t}$$

$$T(0) = T_s = \frac{Q}{\lambda mc} - \frac{Q}{(\lambda-1)mc} + T_s + C$$

$$\Rightarrow C = \frac{Q}{(\lambda-1)mc} - \frac{Q}{\lambda mc} = \frac{Q}{mc}\left[\frac{1}{\lambda-1} - \frac{1}{\lambda}\right]$$

$$\therefore T(t) = \frac{Q}{\lambda mc} - \frac{Q}{(\lambda-1)mc}e^{-t} + T_s + \frac{Q}{mc}\left[\frac{1}{\lambda-1} - \frac{1}{\lambda}\right]e^{-\lambda t}$$

5 (a) let $\det(A-\lambda I)=0$

$$\begin{vmatrix} 1-\lambda & -4 \\ 4 & 9-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)(9-\lambda) - (4)(-4) = 0$$

$$(\lambda-1)(\lambda-9) + 16 = 0$$

$$\lambda^2 - 10\lambda + 25 = 0$$

$$(\lambda-5)^2 = 0$$

$$\therefore \lambda_{1,2} = 5$$

consider $\lambda = \lambda_{1,2} = 5$

$$\left[\begin{array}{cc|c} 1-\lambda & -4 & 0 \\ 4 & 9-\lambda & 0 \end{array}\right] \xrightarrow{\lambda=5} \left[\begin{array}{cc|c} -4 & -4 & 0 \\ 4 & 4 & 0 \end{array}\right] \xrightarrow[r_2=r_2+r_1]{r_1=\frac{1}{4}r_1} \left[\begin{array}{cc|c} 1 & 1 & 0 \\ 0 & 0 & 0 \end{array}\right]$$

$$\Rightarrow \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -x_2 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

eigenvector $v_1 = (-1)\begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$$\left[\begin{array}{cc|c} -4 & -4 & 1 \\ 4 & 4 & -1 \end{array}\right] \xrightarrow[r_2=r_1+r_2]{r_1=\frac{1}{4}r_2} \left[\begin{array}{cc|c} 1 & 1 & -\frac{1}{4} \\ 0 & 0 & 0 \end{array}\right]$$

$$\Rightarrow \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -\frac{1}{4}-x_2 \\ x_2 \end{bmatrix} = \begin{bmatrix} -\frac{1}{4} \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

eigenvector $v_2 = \begin{bmatrix} -\frac{1}{4} \\ 0 \end{bmatrix} + \left(-\frac{1}{4}\right)\begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -\frac{1}{4} \end{bmatrix}$

$\therefore A = PJP^{-1}$ where $P = \begin{bmatrix} 1 & 0 \\ -1 & -\frac{1}{4} \end{bmatrix}$, $J = \begin{bmatrix} 5 & 1 \\ 0 & 5 \end{bmatrix}$

$$P^{-1} = \frac{1}{-\left(-\frac{1}{4}\right)} \begin{bmatrix} -\frac{1}{4} & 0 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ -4 & 4 \end{bmatrix}$$

5(b) $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & -4 \\ 4 & 9 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \quad \begin{bmatrix} x(0) \\ y(0) \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$$A = \begin{bmatrix} 1 & -4 \\ 4 & 9 \end{bmatrix} = PJP^{-1}$$

$$e^{At} = Pe^{Jt}P^{-1}$$

$$= \begin{bmatrix} 1 & 0 \\ -1 & -\frac{1}{4} \end{bmatrix} \begin{bmatrix} e^{5t} & te^{5t} \\ 0 & e^{5t} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -4 & -4 \end{bmatrix}$$

$$= \begin{bmatrix} (1-4t)e^{5t} & -4te^{5t} \\ 4te^{5t} & (1+4t)e^{5t} \end{bmatrix}$$

$$\begin{bmatrix} x(t) \\ y(t) \end{bmatrix} = e^{At} \begin{bmatrix} x(0) \\ y(0) \end{bmatrix}$$

$$= e^{At} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} -4te^{5t} \\ (1+4t)e^{5t} \end{bmatrix}$$